

Falcon Core: Changes from 2017 to present

Overview

This document provides a comprehensive changelog from 2017 to present (Falcon 3.0). This major version includes significant architectural improvements, refactoring, and new features.

New Features

Falcon Workbench (Under Development)

- **Visual pipeline designer** - Design and modify signal processing graphs through interactive workbench instead of manual YAML editing
- **Drag-and-drop interface** - Easily add, connect, and configure processors using intuitive drag-and-drop functionality
- **Processor library** - Browse and search available processors with descriptions and documentation
- **Graph validation** - Real-time validation of graph connections and configurations to prevent errors
- **Pipeline deployment** - Deploy designed pipelines directly to Falcon server instances from the workbench
- **Live monitoring** - Visualize data flow and processor status in real-time during pipeline execution
- **Integrated documentation** - Access processor and system documentation directly within the workbench interface
- **Export and import graphs** - Easily share and reuse signal processing graphs through export/import functionality
- **Keyboard shortcuts** - Extensive keyboard shortcuts for efficient navigation and operation
- **Feedback system** - Detailed GUI logs available for troubleshooting in Github issues and user feedback collection
- **Regular updates** - Update manager to keep the workbench and its components up to date with the latest features and fixes

Documentation System

- **Comprehensive documentation framework** [[c929939](#), [3eb218b](#)]:
 - Set up Doxygen and Sphinx documentation structure
 - Added C++ reference API documentation [[a43cf3a](#)]
 - Added developer documentation [[a43cf3a](#), [7e28874](#)]
 - Processor documentation in table format [[5eb3534](#), [e2b0748](#)]
 - Data type descriptions [[d09931b](#), [a4cbb5](#)]
 - Graph system documentation [[3ff3d9f](#)]
 - Multiple documentation updates [[26f2b92](#), [c1cf97e](#), [1a33105](#), [6a23406](#)]
- **In-code documentation registration** [[37a17bd](#), [23cae0c](#)] - Processors can register documentation
- **Documentation commands** [[04d8086](#), [8b27bb4](#)] - Users can request docs via CLI (docs/d/D commands)
- **Docker installation documentation** [[c68c82d](#)]
- **Documentation organization improvements** [[d3dd1b1](#), [59ed4b9](#), [2dbaeac](#)]

- **Example documentation** - Ripple detection example [[82c6ef3](#)]

CLI & Commands

- **Resources command** [[ff83fe4](#), [58cbcb](#)] - New command to list and manage resources
- **Version information** [[5564c0f](#)] - Added versioning info on request with git tag + build timestamp [[1694827](#)]
- **Extension info in versioning** [[99bc75d](#)] - Track extension versions
- **Improved Falcon CLI** [[8614a95](#)] with better command handling

State Management

- **State alias export enhancements:**
 - Export description and value if state permission is readable [[483110b](#)]
 - Use value and description from alias instead of dependent [[4a69fa2](#)]
 - Separate readable/writable states [[d3f8fa6](#)]
- **Shared state convenience methods** [[70b1d35](#), [2360f3a](#)] - Added convenience shared type state creation
- **Refactored shared state system** [[5b6aa4e](#), [0d2ff64](#)]
- **Static member functions** [[5fc8af2](#)] for data type and data name string retrieval

Validation & Error Handling

- **Validated options system** [[142411c](#), [4c6a717](#)]:
 - Added `ValueMap` [[ddabb0f](#)] for validated values
 - Measurement values with units support [[0c5e6c9](#)]
 - Custom units defined in main.cpp [[0c5e6c9](#)]
 - File and directory validation via filesystem library [[e179961](#)]
 - Range class with `to_string` method [[447ae9f](#)]
- **Improved error messages:**
 - Better error messages in `parseConnectionRule` [[fd47b19](#)]
 - Show existing value name in error messages [[4498d30](#)]
 - Raise errors for unknown options [[74f87e6](#), [6f5969b](#)]
 - Catch bad documentation format [[1cafdab](#)]
 - Add error messages for resources [[c2a67a3](#)]

Serialization

- **Serialization Encoding enum** [[b0b4523](#)] added
- **YAML conversion support** [[b0b4523](#)] for serialization enums
- **Common serialization code** [[9cb24d8](#)] extracted to separate file
- **FlatBuffers/FlexBuffers integration** [[30e24d4](#)] for high-performance serialization

Flexibility & Usability

- **Flexible naming syntax:**
 - Variable name flexibility for state names [[2137a65](#)]
 - Equivalent separators: space, dash, underscore [[2137a65](#)]
 - Flexible processor name syntax in configuration [[e626372](#), [cea133d](#)]

- Flexible port name parsing [[8a410aa](#)]
 - **Multi-processor specification** [[de8f790](#), [6984fee](#)] improvements
 - **Stream name** [[bd208ea](#)] added to StreamInfo object
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Breaking Changes

Data Type System Overhaul

- **Major refactoring of data type system** [[516d44c](#), [a3af8fa](#)] - Complete redesign of how data types are handled
- **Separated data type class from data payload class** [[2e83ad8](#)] - Data types and payloads are now distinct entities
- **Removed `Initialize` methods on data classes** [[433123f](#)] - Initialization now done through constructors with new ringbuffer implementation
- **Changed data type serialization format** - Now uses FlatBuffers/FlexBuffers for improved performance:
 - Schema version 2 introduced [[f8bfeea](#)]
 - Added type information to FlatBuffers format [[0423e29](#)]
 - Stream data format updated to use FlexBuffers [[e7c640c](#), [7c6af80](#)]
 - FlatBuffer serialization implementation [[c12ce20](#), [30e24d4](#)]

API Changes

- **Merged `IProcessor` and `ProcessorEngine` classes** [[f1cbe60](#)] - Simplified processor architecture
- **Removed `StreamConnection` classes** [[ccda54f](#)] - Ports/slots now have direct pointers to their parents
- **Changed state creation API:**
 - Removed `create state` overload with option name [[5a200b5](#)]
 - State creation now requires options lookup; error if option not found [[70dce44](#), [9882488](#)]
 - Default external permission for static states changed to write [[70dce44](#)]
- **Removed flexible format for state names** [[0d268c3](#), [2665979](#)] - Standardized naming convention
- **Changed graph configuration format** [[cda40a3](#), [4e52493](#)] - New YAML structure (manages both versions with warnings [[4a1f499](#)])

Build System

- **Moved from C++11 to C++17** [[18cf842](#)] - Requires modern compiler support
- **Changed CMake structure** [[47062f2](#), [b9f8305](#)] - Extensions now in separate repository
- **New extension management** [[e6bf295](#)] - Uses `FetchContent` for dependencies
- **Upgraded dependencies** [[041cf85](#)]:
 - ZMQ updated - removed deprecated warnings [[d3d64cf](#)]
 - yaml-cpp updated to >0.6 - no longer requires Boost [[e6bf295](#)]
 - Added LLNL/units library dependency [[d95865b](#)]
 - Added FlatBuffers library [[ea90714](#), [c69b5b5](#)]

Configuration

- **Refactored configuration system** [[ddd50c5](#), [73e3c49](#), [b9cbcd3](#)] - New options-based configuration with validation

- **Changed default resource location** [[ddd50c5](#)] to `./` from previous paths
 - **Config port changed** [[9cf46a3](#)] - Fixed to 5555 when creating socket (previously configurable but broken)
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Improvements

Performance & Code Quality

- **Performance lints** [[fb84d44](#)] applied across codebase (#133)
- **Integrated clang-tidy** [[6e19f5c](#)] for static analysis (#132)
- **Code formatting:**
 - Formatted entire codebase [[1c6fb83](#), [18644b2](#), [cea041d](#)]
 - Added clang-format CI [[32a4030](#)]
 - cplint code cleaning [[23554ba](#)]
- **Memory leak fixes:**
 - Fixed memory leak due to missing virtual destructor [[17bb8ba](#)] (#60)
 - Core extension memory leak hotfix [[1538cf1](#)]
 - Removed allocation/deallocation of FlatBuffer builder [[a6c0f9b](#)]
- **Code cleanup** [[13141f3](#), [b230973](#), [d646f67](#)]

Build System & CI

- **Build and clang-format CI** [[32a4030](#), [a897171](#)] (#120, #121)
- **CPP build CI** [[5075944](#)] added
- **Debug/Release build separation** [[7f180f4](#)] - Different build configurations
- **Extension development mode** [[9b2fb3f](#)] added
- **Simplified extension enabling** [[e70a6a3](#), [40c8e62](#)] - Extensions specified in text file, handled by CMake
- **Extension parser** [[06c07c1](#)] moved to separate CMake script (#127)
- **Post-build resource handling** [[cbe2e9e](#)] - Resources copied after build/install
- **CMake improvements** [[486503a](#), [fb6641e](#)]
- **Extension version defaults** [[e10e18e](#)] - Bump to 1.0.0 by default

Logging

- **Upgraded to g3log library** [[ce66411](#)] - Complete logging system overhaul
- **Exported logging implementation** [[29652c8](#)] as library
- **Color support** [[186795e](#)] in console logs
- **Changed default logging path** [[6700fc4](#)] to follow filesystem hierarchy standard
- **Removed unused event levels** [[7161c74](#)]
- **Added log to print graph file path** [[5a13c9f](#)] (#129)
- **Fixed shared library error** [[dbcde54](#)]

Architecture & Refactoring

- **Refactored slot class:**
 - Created `FlushData` method [[b7f38ee](#)]
 - Removed timeout as property, added as option in `retrieveData` methods [[b7f38ee](#)]

- Added docstrings [[e432db8](#), [1075b7c](#), [2edc9fe](#)]
- **StreamInfo class** [[c4f1f39](#)] separated into own file
- **String utilities** [[c7412f4](#), [1cafdab](#)] added for common operations
- **Enhanced port/slot addressing** [[9151689](#)] with processor class and datatype information
- **Port compatibility verification** [[9151689](#)] respects data type hierarchy
- **Graph management improvements:**
 - Transform server-side graph to user-side graph on request [[430bc24](#), [8122acf](#)]
 - Loop on saved YAML then expand processor names [[28aab03](#)]
 - Resource type for resource list command [[c2a67a3](#)]
- **Value class improvements** [[c7412f4](#), [2be9ed2](#), [fd47b19](#)] - YAML conversion and bug fixes
- **Template value checking** [[61bf8a3](#)]
- **Type string utilities** [[6c8bff5](#)] - Added char overload
- **Moved common configuration code** [[d231185](#)] to utilities lib for reuse

Configuration & Resources

- **Resource management:**
 - Combine resources from all extensions [[baa72e1](#)]
 - Copy resources to installation path with `make install` [[6c61a14](#)]
 - Move resources to /share folder [[66c743c](#)]
 - Added YAML extensions support [[a6e047a](#)]
 - Fixed resource path bugs [[36838ae](#), [b889c0a](#), [8c7643e](#), [cdec159](#)]
- **Improved configuration loading:**
 - Filesystem support with validation [[ddd50c5](#)]
 - Auto-create default config path if missing [[ddd50c5](#)]
 - Both graph format versions supported (with warnings) [[4a1f499](#)]
- **Parse build command inputs** [[e23b1a3](#)] - Accept filename or string graph description
- **Option name conversion** [[475035d](#)]
- **Pass by reference optimizations** [[5af317f](#), [392999f](#)]

Bug Fixes

- **Boolean shared state** [[59cef09](#)] - Fixed not loading correctly with true/false strings
- **Boolean YAML serialization** [[fc820d9](#)] - Replace 0/1 with true/false
- **Multichannelfilter crash** [[eb3a1e2](#)] - Fixed crashing when unprepare
- **Remote graph path** [[49ec497](#)] - Fixed not resolving when sent from client
- **Connection parser** [[ae41cba](#)] - Fixed for slot part
- **Parsing bugs** [[de8f790](#), [cea133d](#)] - Fixed in processor name and connection rule parsing
- **ZMQ communication** [[78d0aa4](#)] - Fixed instability after upgrade
- **Stream port timeout** [[6c11afd](#)] - Fixed uint instead of int
- **Trailing space/newline issues** [[2ee4681](#), [0b4d22d](#), [04c5644](#)]
- **IData parent class call** [[d13c0f9](#)]
- **Template compilation** [[3d3ca26](#)] - Added constexpr is_convertible check
- **Compilation errors** [[5dff4ae](#), [c989a86](#)] - Fixed various issues
- **CleanData typo** [[320e708](#)] - Restored method removed by error
- **Regex for options** [[f8b2341](#)] - Removed as it doesn't always work
- **Exception handling** [[0db00a3](#)] - Throw exception one level upper

- **Small bug fixes** [[1a63d63](#), [44f8b3a](#)]
 - **Typo fixes** [[e1193da](#)]
 - **Extension updates** [[a773c2e](#), [2c9e058](#), [6e1d120](#)] - Point to correct versions
 - **Debug logging** [[7d6becb](#)] - Removed log messages
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Dependencies & Libraries

Added

- **FlatBuffers library** [[ea90714](#), [c69b5b5](#)] for serialization
- **LLNL/units library** [[d95865b](#)] for measurement values
- **Filesystem library** [[e179961](#)] support
- **g3log** [[ce66411](#)] for logging

Updated

- **ZMQ** [[d3d64cf](#)] - Removed deprecated API usage
- **yaml-cpp** [[e6bf295](#)] to >0.6 - Removed Boost dependency
- **Core extension versions** - Updated throughout development [[1538cf1](#), [2c9e058](#), [6e1d120](#)]
- **Dependencies upgrade** [[041cf85](#)]

Removed

- **Boost library dependency** [[e6bf295](#)] - No longer needed with yaml-cpp >0.6
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Project Structure Changes

- **Extensions moved to separate repository** [[47062f2](#), [b9f8305](#)]
 - **Resources organized in /share folder** [[66c743c](#)] following standard hierarchy
 - **Logging path follows filesystem hierarchy standard** [[6700fc4](#)]
 - **Install path restructured** for better organization
 - **Extensions managed via FetchContent** [[e6bf295](#)] CMake module
 - **Extension files in .gitignore** [[120b1a4](#)]
 - **Keep only open-source extensions** [[4facbfd](#)] in release
 - **Exclude data types and processors** [[395ac7c](#)] from compilation (optional)
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Documentation Improvements

- **Created CHANGELOG.md** [[1be23e1](#)] (#126)
- **Comprehensive manual documentation** [[9429f31](#), [adadd3f](#), [cd6761e](#), [c8c3f01](#)]
- **Ripple detection example** [[82c6ef3](#)] added
- **Processor categories reorganized** [[59ed4b9](#)]
- **Readable processor descriptions** [[e2b0748](#), [b455536](#)]
- **Installation guide updated** [[422d31b](#), [b889c0a](#)]
- **Client building documentation** [[9c0fcf1](#)]
- **README updates** [[488aa93](#), [4c9c6b6](#)]

- **Documentation updates** [[17a25a3](#), [49b8a46](#), [2367fb9](#), [bbb1b27](#)]
- **Doc build requirements** [[7f00c16](#), [4f9588c](#)]
- **Documentation index updates** [[0aa9e03](#), [14a3308](#)]
- **Remove extension-specific documentation** [[8002e8b](#)]
- **Documentation fixes** [[636fcfb](#), [197095b](#)]
- **Method name changes** [[0c230d0](#)]
- **Remove shortcut doc** [[b235ae7](#)] - Add documentation at registration